

Weekly Progress Report

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Domain: Data Science and Machine Learning

Date of Submission: 11/07/2026

Week Ending: 03

I. Overview:

This week focused on understanding the Smart City Traffic Forecasting project requirements, exploring the traffic dataset, performing data preprocessing, feature engineering, and building the initial machine learning pipeline for traffic volume prediction.

II. Achievements:

1. Project Understanding:

Project Name: Smart City Traffic Forecasting using Machine Learning

- Understood the objective of predicting traffic volume using historical traffic data.
 - Studied the provided traffic dataset containing DateTime, Junction and Vehicle information.
 - Explored the project requirements, workflow and dataset structure.
 - Identified the target variable and input features required for model training.
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2. Data Preparation:

- Organized training and testing datasets.
- Implemented data loading pipeline.
- Performed data cleaning and preprocessing.
- Handled missing values and duplicate records.
- Converted DateTime column into proper datetime format.

- Prepared dataset for feature engineering.
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3. Feature Engineering:

Implemented various feature engineering techniques including:

- Year
 - Month
 - Day
 - Hour
 - Minute
 - Day of Week
 - Quarter
 - Weekend Indicator
 - Season Identification
 - Morning Rush Hour Detection
 - Evening Rush Hour Detection
 - Business Hour Detection
 - Night Hour Detection
 - Cyclic Encoding using Sine and Cosine Transformation
 - Month Boundary Indicators
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4. Machine Learning Model Development:

Successfully developed the initial machine learning pipeline.

- Implemented Random Forest Regressor.
- Performed Train-Test Split.
- Trained the prediction model.
- Generated traffic predictions.

- Evaluated model performance using:
 - Mean Absolute Error (MAE)
 - Root Mean Squared Error (RMSE)
 - R^2 Score
 - Saved the trained model using Joblib.
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III. Challenges:

1. Dataset Understanding:

- Understanding traffic flow patterns.
 - Selecting meaningful temporal features.
 - Handling DateTime information correctly.
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2. Feature Engineering:

- Creating effective time-based features.
 - Implementing cyclic encoding for periodic data.
 - Identifying peak traffic hours for better prediction accuracy.
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IV. Learning Resources:

- Scikit-Learn Documentation
 - Pandas Documentation
 - NumPy Documentation
 - Matplotlib Documentation
 - Plotly Documentation
 - Streamlit Documentation
 - Random Forest Regression Tutorials
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V. Next Week Goals:

- Develop interactive Streamlit Dashboard.
 - Implement CSV upload functionality.
 - Add Traffic Prediction module.
 - Build Feature Importance visualization.
 - Add Correlation Heatmap.
 - Implement Prediction Distribution chart.
 - Add Model Metrics dashboard.
 - Improve User Interface.
 - Prepare project documentation.
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VI. Additional Comments:

“Successfully completed data preprocessing, feature engineering, and machine learning model development. The traffic prediction model was trained successfully and achieved good prediction performance. The project is ready for Streamlit dashboard development and visualization phase.”